

Session: Data Preprocessing

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Applied Machine Learning

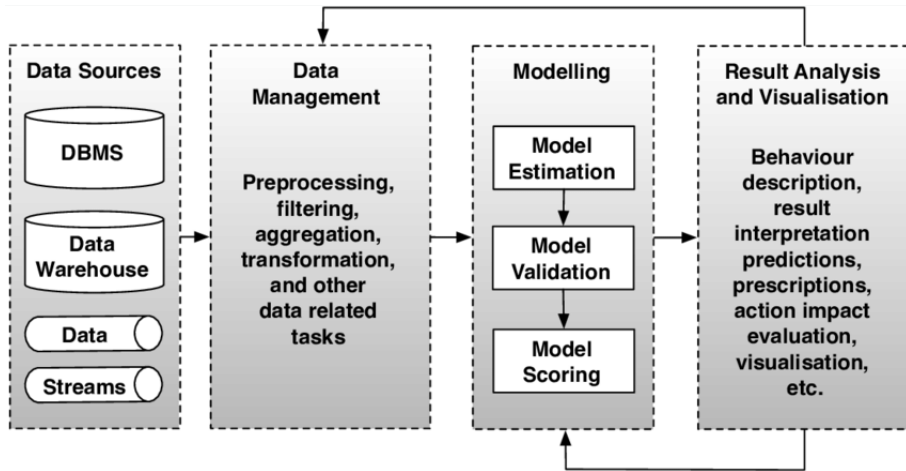
Master in Data Science and Analytics



Strathmore University

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ML Workflow



Summary

Data Objects and Attribute Type

Basic Statistical Description of Data

Data Visualization

Data Preprocessing

Data Objects

Data Point

- A **data point** represents an entity.
 - Data points are described by feature/attributes.
 - Also called, example, input instance, or simply point.
- **Feature** (or dimensions or attributes)
 - A field representing state of nature or characteristic of a data point.
- Example: a patient data point with 4 features

	Gender	Age	Condition 1	Condition 2
x_i	Male	53	Heart Attack	HIV

Data Objects

Data Set

- A **dataset** is made up of data points.
- Example:
 - Patient dataset containing information of 100 patients.
 - Dog dataset containing information of 40 dogs.

Data Objects

Data Matrix

- Mathematically, a data is a vector (point) in a multidimensional space, $x_i = \{x_i^1, \dots, x_i^l\}$ where l is dimensionality of the data.
- So, a **dataset** of n points is a matrix of $N \times l$

$$\begin{bmatrix} x_1^1 & x_1^2 & \dots & x_1^l \\ x_2^1 & x_2^2 & \dots & x_2^l \\ \vdots & \vdots & \ddots & \vdots \\ x_n^1 & x_n^2 & \dots & x_n^l \end{bmatrix}$$

- In Google Tensorflow, a **Tensor** is a multi-dimensional data structure $d = 0, 1, 2, 3, \dots$

Attribute Types

I

- **Nominal (categorical):** categories, states, or names of things
 - HairColor = (black, blond, brown, grey, red, white)
 - Marital status, occupation
- **Binary:** nominal attribute with only 2 states (0 and 1)
 - *Symmetric binary:* both outcomes equally important (e.g., gender)
 - *Asymmetric binary:* outcomes not equally important. (e.g., medical test: positive vs. negative)

Attribute Types

II

- **Ordinal:**
 - Values have a meaningful order (ranking) but magnitude between successive values is not known.
 - Size = (small, medium, large), grades, army rankings
- **Numeric:**
 - Quantity (integer or real-valued)

Data Objects

Special Data Types

- **Time Series** : sensors data, stock market value, predictions, ...
 - Data has a time dependency
 - Online analysis based on correlations (p.e. search markers in stock market values, ...)
 - Offline analysis based on shapes (p.e. clusterization of ECGs)
- **Sequential Data**: Biological sequences, weblogs
- **Stream**:
 - Real-time, one-step algorithms
 - Create efficient intermediate representations
- **Text**:
 - Features: high dimensionality, sparse, values corresponds to word frequencies
- **Multimedia Data**: Images, audio and video

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Basic Statistical Description of Data

- **Motivation**
→ To better understand the data: central tendency, variation and spread.
- **Central tendency**
→ Mean, median, mode
- **Variation and spread**
→ Variance, standard variation, quantile.

Basic Statistical Description of Data

Measuring the Central Tendency

- **Mean**

- Arithmetic mean: $\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$
- Weighted arithmetic mean: $\bar{x} = \frac{\sum_{i=1}^N w_i x_i}{\sum_{i=1}^N w_i}$
- Trimmed mean: chopping extreme values

- **Median**

- Sort the dataset in an increasing order
- Take the middle value if there is odd number of data points
- Take an average of the middle two values if there is even number of data points

- **Mode**

- Value that occurs most frequently in the data
- A dataset can be unimodal (1 mode), bimodal (2 modes), etc.

Basic Statistical Description of Data

Measuring dispersion of data: Variance and standard deviation

- **Variance:** $\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2$
- **Standard deviation:** σ is the square root of variance
- **Quantile:** set of P points that divide the range of probability distribution into $P - 1$ intervals (with equal probability density)
 - $P=2$: 2-Quantile = Median
 - $P=4$: 4-Quantile = Quartile
 - $P=100$: 100-Quantile = Percentile

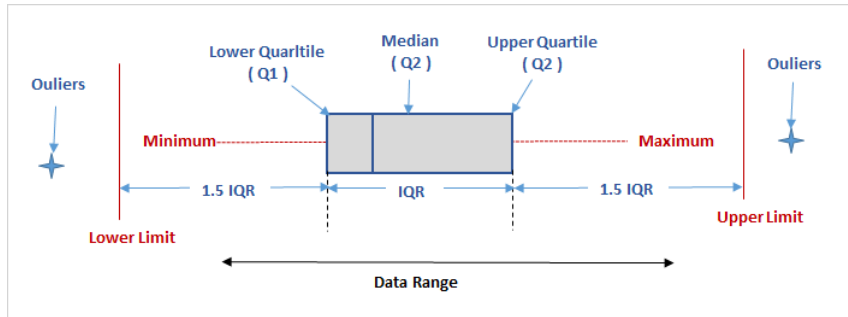
Basic Statistical Description of Data

Measuring dispersion of data: Five Number Summary

- Inter-quartile range: $IQR = Q3 - Q1$
- Five numbers summary: min, Q1, median, Q3, max
- **Outlier**: usually, a value greater than $Q3 + 1.5 \times IQR$ or less than $Q1 - 1.5 \times IQR$

Basic Statistical Description of Data

Five Number Summary



Summary

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Basic Statistical Description of Data

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Data Visualization

- Why data visualisation?
 - Gain insight into an information space by mapping data onto graphical primitives
 - Provide qualitative overview of large data sets
 - Help find interesting regions and suitable parameters for further quantitative analysis
- There are numerous methods:
 - **Basic statistical visualisation techniques**
 - Icon-based visualisation techniques
 - Hierarchical visualisation techniques

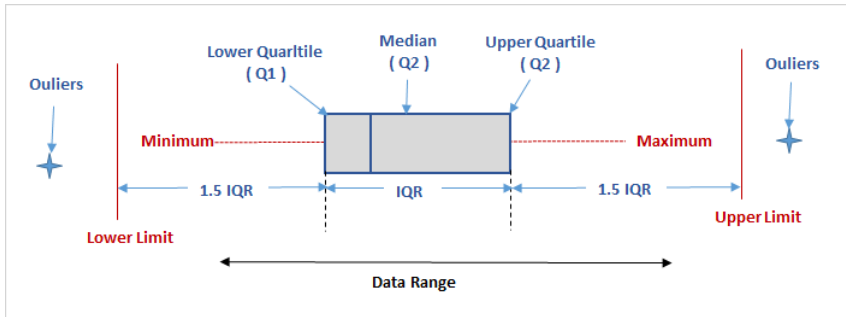
Data Visualization

Basic Statistical Visualisation

- **Boxplot:** graphic display of five-number summary
- **Histogram:** x-axis are values, y-axis represents frequencies
- **Scatter plot:** each pair of values is a pair of coordinates and plotted as points in the plane

Basic Statistical Visualisation

Boxplot



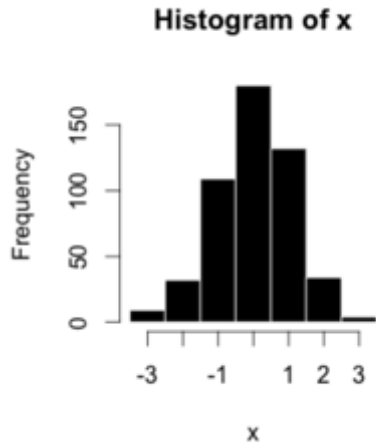
Basic Statistical Visualisation

Histogram

- Histogram: Graph display of tabulated frequencies, shown as bars
- It shows what proportion of cases fall into each of bars
- The categories are usually specified as non-overlapping intervals of some variable. The categories (bars) must be adjacent

Basic Statistical Visualisation

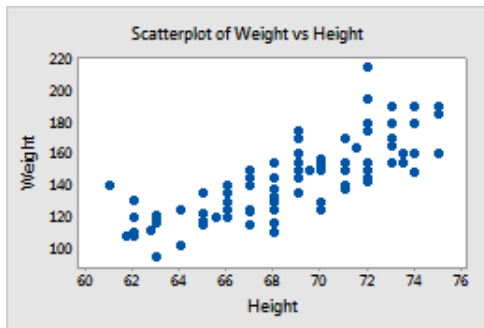
Histogram



Basic Statistical Visualisation

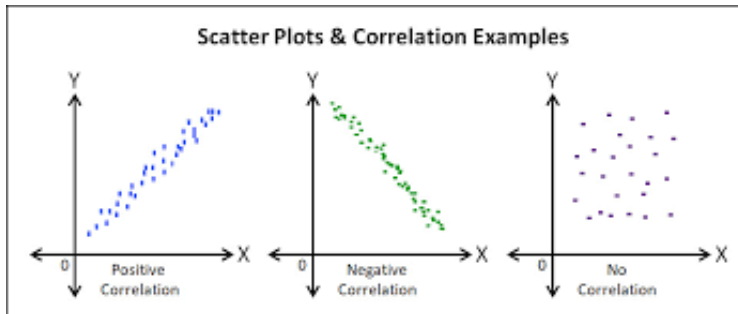
Scatter Plot

- Provides a first look at bivariate data to see clusters of points, outliers, etc
- Each pair of values is treated as a pair of coordinates and plotted as points in the plane



Basic Statistical Visualisation

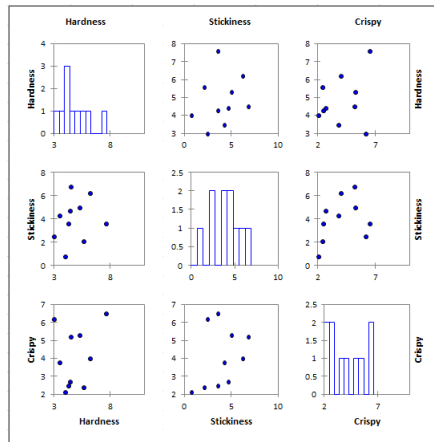
Examples of Scatter Plot



Basic Statistical Visualisation

Scatter plot for high dimensional data

- Projection data onto lower dimension, i.e. 2-D
- Do scatter plot for every pair of dimensions in turns



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Why Preprocess the Data?

Measures for data quality

- **Accuracy:** correct or wrong, accurate or not
- **Completeness:** not recorded, unavailable, ...
- **Consistency:** some modified but some not, dangling, ...
- **Timeliness:** timely update

Major Tasks in Data Preprocessing

- **Data cleaning**
 - Fill in missing values
 - Smooth noisy data
 - Identify or remove outliers
- **Data integration**
 - Integration of multiple data sources (databases)
 - Resolve inconsistencies
- **Data reduction:** Dimensionality reduction
- **Data transformation and data discretisation:** Normalisation, Standardisation

Data Cleaning

Why data cleaning?

- Data in the Real World Is Dirty
- Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, transmission error
- Two major dirtiness in data
 - **Incomplete (missing)**: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
⇒ e.g., Occupation = " " (missing data)
 - **Incorrect (noise)**: containing noise, errors, or outliers
⇒ e.g., Salary = "-10" (an error)

Data Cleaning

Incomplete Data

- Data is **not always available**. E.g., many tuples have no recorded value for several attributes, such as customer income in sales data
- Missing data may be due to
 - equipment malfunction
 - inconsistent with other recorded data and thus deleted
 - data not entered due to misunderstanding
 - certain data may not be considered important at the time of entry
 - not register history or changes of the data
- Missing data may need to be inferred

Data Cleaning

How to Handle Missing Data?

- Ignore the data point: usually done when class label is missing (when doing classification)—not effective why?
- Fill in the missing value manually: problem?
- Fill in automatically with
 - a global constant : e.g., “unknown”, “0”.
 - the attribute mean
 - the attribute mean for all samples belonging to the same class: smarter

Data Cleaning

Incorrect (Noisy) Data

- **Noise:** random error or variance in a measured variable
- **Incorrect attribute values** may be due to
 - Faulty data collection instruments
 - Disagreement between data experts
 - Data transmission problems
 - Technology limitation

Data Cleaning

How to Handle Noisy Data?

- **Binning:** Used to reduce the effects of minor observation error by consulting neighbours.
- **Clustering:** Detect and remove outliers
- **Regression:** Smooth by fitting the data into regression functions
- **Combined computer and human inspection:** Detect suspicious values and check by human (e.g., deal with possible outliers)

Data Integration

- Combines data from multiple sources into a coherent store
- Schema integration: e.g., $A.cust-id \equiv B.cust-num$
⇒ Integrate metadata from different sources
- Entity identification problem:
⇒ Identify real world entities from multiple data sources, e.g., Bill Clinton = William Clinton (same person)
- Detecting and resolving data value conflicts
 - For the same real world entity, attribute values from different sources are different
 - Possible reasons: different representations, different scales, e.g., metric vs. British units

Data Reduction

Strategies

- **Data reduction:** Obtain a reduced representation of the data set that is much smaller in volume but yet produces the same (or almost the same) analytical results
- **Why data reduction?** — A database/data warehouse may store terabytes of data. Complex data analysis may take a very long time to run on the complete data set.
- Data reduction strategies:
 - **Dimensionality reduction**, e.g., remove unimportant features
 - Principal Components Analysis (PCA)
 - Feature subset selection
 - **Data reduction**
 - Histograms
 - Clustering
 - Sampling
 - Model-based (Gaussian Mixture Model)

Data Transformation

Methods

A function that maps the entire set of values of a given attribute to a new set of replacement values s.t. each old value can be identified with one of the new values

- Attribute/feature construction
 - New attributes constructed from the given ones
 - Image $\Rightarrow$ data point, Signal $\Rightarrow$ data point
- Normalization: Scaled to fall within a smaller, specified range
 - min-max normalisation (0,1)
 - z-score normalisation (μ : mean, σ : std dev: a.k.a standardisation)
- Discretisation: Transform continuous variable to discrete ones.

Data Preprocessing

Summary

- **Data quality:**
⇒ accuracy, completeness, consistency, timeliness
- **Data cleaning:** e.g. missing/noisy values, outliers
- **Data integration** from multiple sources:
⇒ Remove redundancies; Detect inconsistencies
- **Data reduction** (Not so important now)
⇒ Dimensionality reduction; Dataset size reduction
- **Data transformation and data discretization**

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References

- Aurelien Geron, *Hand-on Machine Learnings with Scikit-Learn, Keras & Tensorflow*, 3th Edition. O'Really. 2022 (Pages 67-87)